

C Programs for Problems 11 to 15

11. Swapping Two Numbers Using Pointers

```
#include <stdio.h>

void swap(int *a, int *b) {
    int temp = *a;
    *a = *b;
    *b = temp;
}

int main() {
    int x, y;

    printf("Enter two numbers: ");

    scanf("%d %d", &x, &y);

    printf("Before swapping: x = %d, y = %d\n", x, y);

    swap(&x, &y);

    printf("After swapping: x = %d, y = %d", x, y);

    return 0;
}
```

12. Count Vowels, Consonants, Digits, and Spaces

```
#include <stdio.h>

#include <ctype.h>

int main() {
    char str[100];
```

```

int vowels = 0, consonants = 0, digits = 0, spaces = 0;

printf("Enter a sentence: ");

gets(str);

for (int i = 0; str[i] != '\0'; i++) {

    if (isalpha(str[i])) {

        if (strchr("AEIOUaeiou", str[i]))

            vowels++;

        else

            consonants++;

    } else if (isdigit(str[i])) {

        digits++;

    } else if (isspace(str[i])) {

        spaces++;

    }

}

printf("Vowels: %d\nConsonants: %d\nDigits: %d\nSpaces: %d", vowels, consonants,
digits, spaces);

return 0;

}

```

13. Binary Search Implementation

```

#include <stdio.h>

int binarySearch(int arr[], int n, int key) {

    int low = 0, high = n - 1;

    while (low <= high) {

```

```

        int mid = (low + high) / 2;

        if (arr[mid] == key)

            return mid;

        else if (arr[mid] < key)

            low = mid + 1;

        else

            high = mid - 1;

    }

    return -1;

}

int main() {

    int n, arr[100], key, result;

    printf("Enter the size of the array: ");

    scanf("%d", &n);

    printf("Enter the sorted array elements: ");

    for (int i = 0; i < n; i++) {

        scanf("%d", &arr[i]);

    }

    printf("Enter the number to search: ");

    scanf("%d", &key);

    result = binarySearch(arr, n, key);

    if (result != -1)

        printf("Number found at index %d", result);

    else

        printf("Number not found");

    return 0;

}

```

14. Check if a Number is Strong

```
#include <stdio.h>

int factorial(int n) {
    int fact = 1;
    for (int i = 1; i <= n; i++) {
        fact *= i;
    }
    return fact;
}

int isStrong(int num) {
    int sum = 0, temp = num;
    while (temp > 0) {
        sum += factorial(temp % 10);
        temp /= 10;
    }
    return sum == num;
}

int main() {
    int num;
    printf("Enter a number: ");
    scanf("%d", &num);
    if (isStrong(num))
        printf("%d is a Strong number", num);
    else
```

```
        printf("%d is not a Strong number", num);

    return 0;

}
```

15. Pattern Printing

```
#include <stdio.h>

int main() {

    int rows;

    printf("Enter the number of rows: ");

    scanf("%d", &rows);

    for (int i = 1; i <= rows; i++) {

        for (int j = 1; j <= rows - i; j++) {

            printf(" ");

        }

        for (int k = 1; k <= (2 * i - 1); k++) {

            printf("*");

        }

        printf("\n");

    }

    return 0;

}
```